

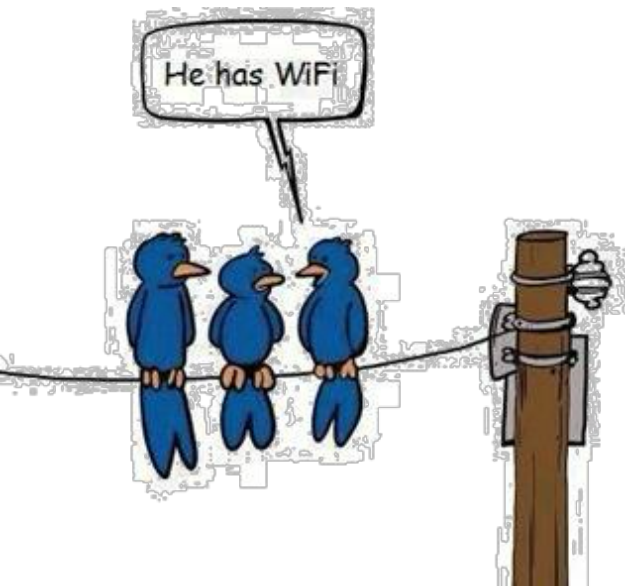
ECSE 4560/ 6560

Modern Communication Systems

Lecture 4: Wireless Systems Overview

Part 3 - Estimation

Ish Jain, Fall 2025



- Institute Advancement film crew
- Video content to promote a fundraising campaign
- Capture footage of Modern Communication Systems class
- Need about 15 minute
- Consent?



Imaging and Sensing for Next-generation Wireless Networks

Ashutosh Sabharwal

Ernest D. Butcher Professor in the Electrical and
Computer Engineering
Rice University

Mercer Distinguished Lecture Series
CII 3051
Wed, September 17, 2025 at 4:00 PM

As the wireless standards occupy larger parts of the spectrum with an ever-larger number of antennas, there is an opportunity to also use the communications spectrum for “imaging” the environment. Thus, there is an opportunity for next-generation networks like 6G to be multi-function, i.e., use the same spectrum for multiple functions. In this talk, we will show how multipath propagation can be harnessed to improve radar-based sensing to (i) image beyond the radar's field of view and (ii) estimate the full velocity vector using multipath reflections. We will also touch upon our recent work on converting base stations into weather radars.

Ashutosh Sabharwal is the Ernest D. Butcher Professor in the Electrical and Computer Engineering department at Rice University. He works in two independent areas - wireless networks and digital health. His ongoing work in wireless focuses on theory and methods for joint wireless communications & imaging, as well as large-scale experimental platforms. He was one of the inventors of in-band full-duplex communications, a technology used in wireline and wireless standards. He founded the WARP project (warp.rice.edu), an open-source project used by 150+ research groups worldwide. He also led Rice RENEW (renew-wireless.org), which developed an open-source software-defined massive MIMO wireless network platform. Concurrent with his wireless research, he also leads the Rice Digital Health Initiative (dhi.rice.edu) and is the Co-director of a new Methodist-Rice Digital Health Institute. His digital health research focus is the development of devices and data science to quantify behavior-biology pathways across many diseases. He received the 2017 IEEE Jack Neubauer Memorial Award, the 2018 IEEE Advances in Communications Award, the 2019, 2021, and 2025 ACM Sigmobility Test-of-time Awards, the 2019 ACM MobiCom Community Contribution Award, and the 2023 ICC Best Paper Award. His research has led to multiple startups and FDA-approved products. He is a Fellow of IEEE, ACM, and the National Academy of Inventors.



Announcements

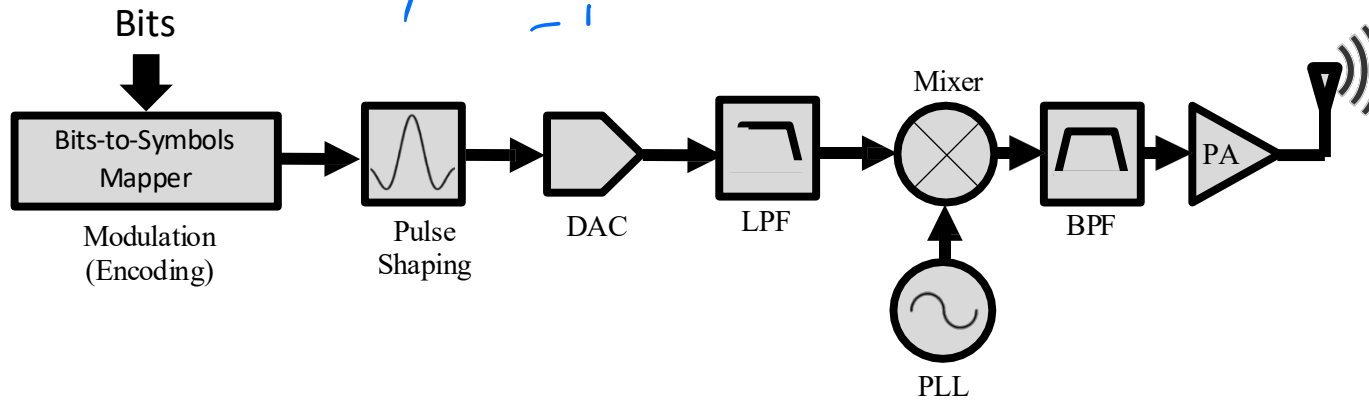
- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- GradeScope for assignment submission (You will be added)
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Estimation**
- Projects
 - A list of papers will be posted
 - Group formation [MCS_Fall25_project.xlsx](#) due 9/12
- HW1/Lab1 will be posted today, due Friday 9/19

Today's Class: Wireless Overview 3

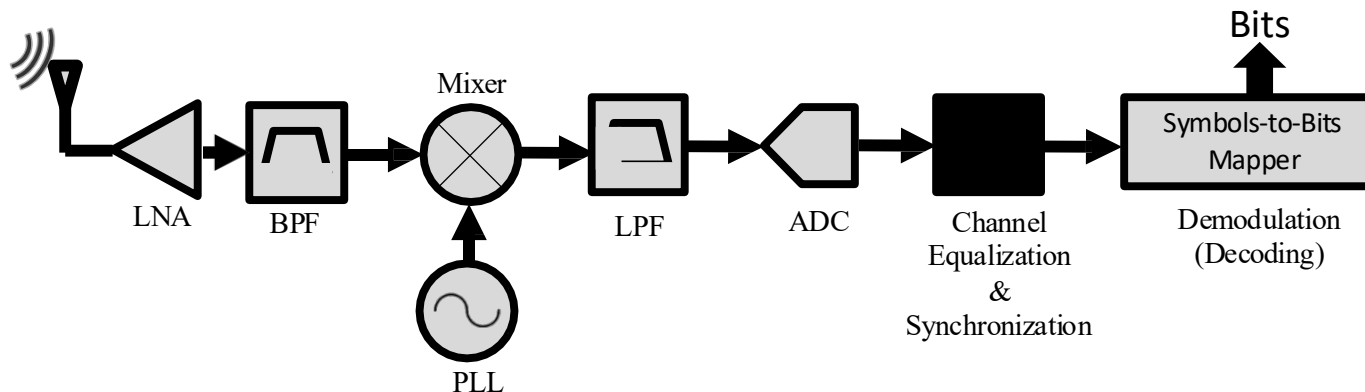
- Decoding under noise (revise)
- MLE and MAP estimators (revise)
- Decision Boundary (revise)
- Bit error rate and symbol error rate
- ADC/DAC Sampling and reconstruction (to continue..)

Wireless Communication System

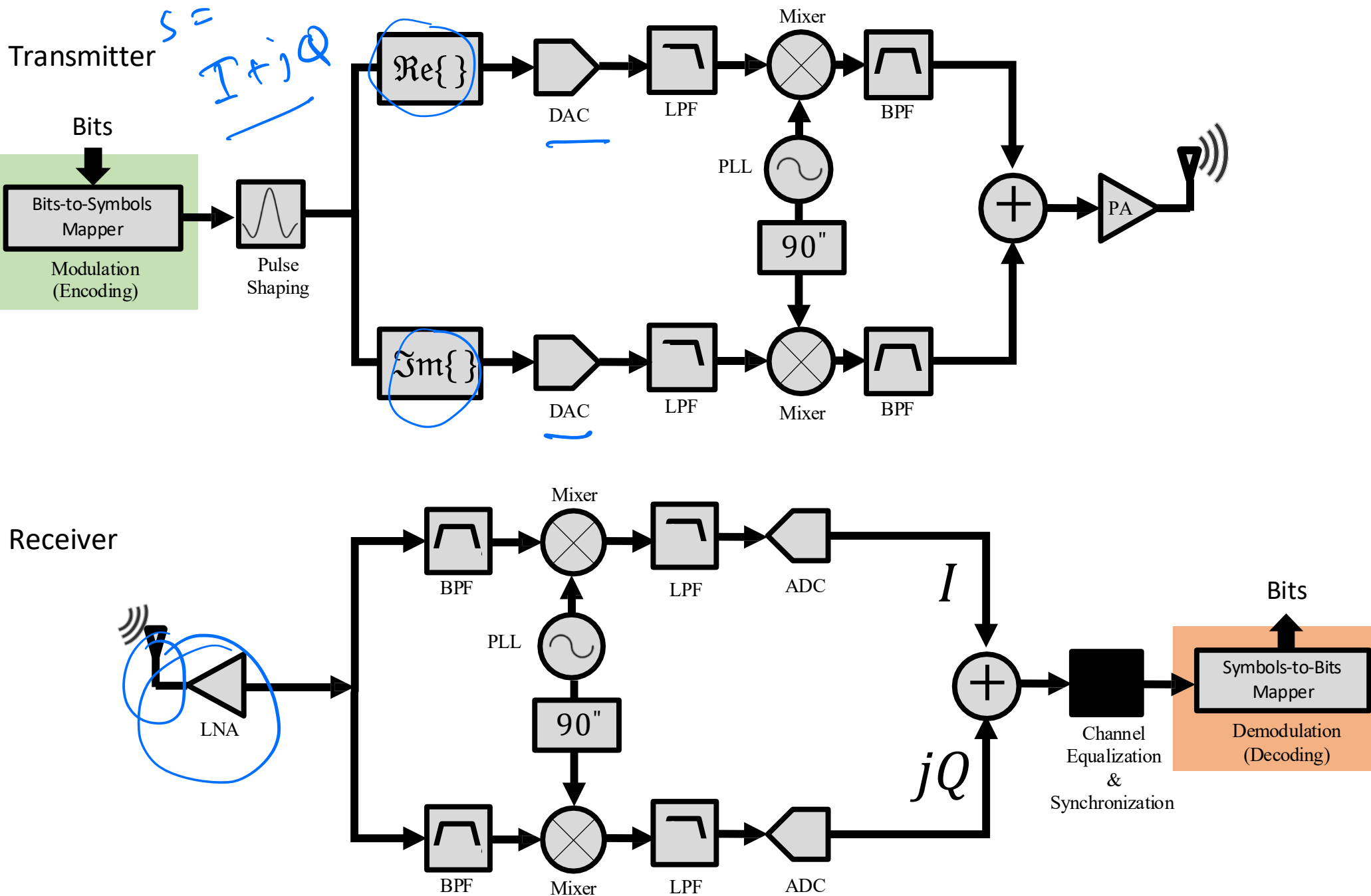
Transmitter



Receiver

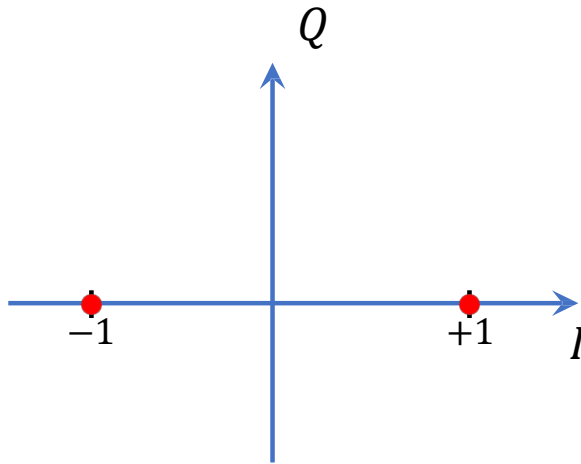


Wireless Communication System



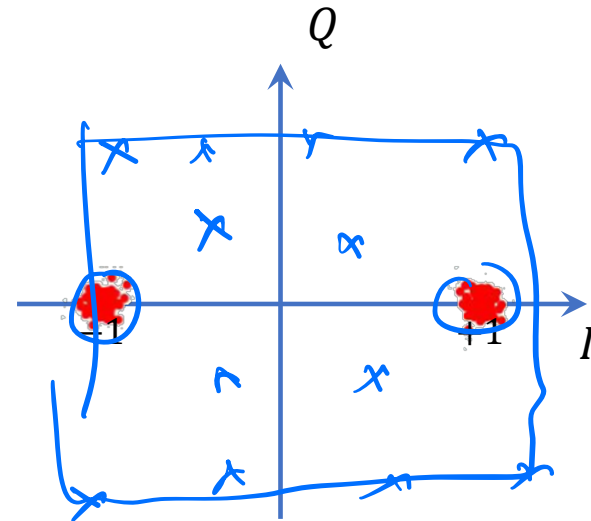
Decoding under noise

$$\begin{aligned} 0 &\rightarrow -1 \\ 1 &\rightarrow +1 \end{aligned}$$



$x(t)$

Transmitted Constellation



$$y(t) = x(t) + v(t)$$

Received Constellation

BPSK
QPSK
16QAM

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}}$$

$$= \frac{E[|x(t)|^2]}{E[|v(t)|^2]}$$

$$= \frac{E_s}{N_0}$$

Noise Power

Mathematical Expression:

$$N_0 = kBT \text{ Watt}$$

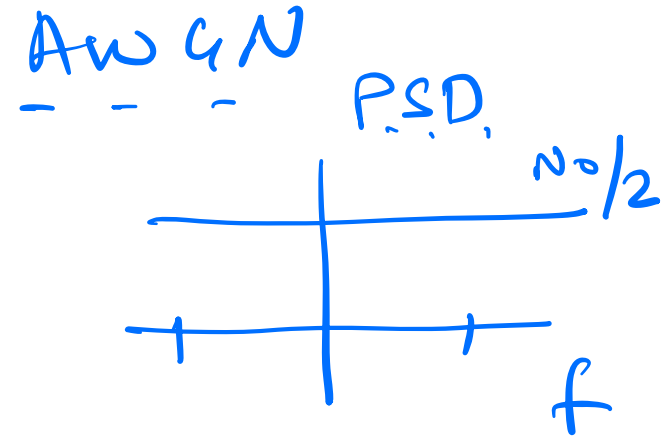
$$P_n = 10 \log_{10} \left(\frac{kBT}{1 \text{ mW}} \right) \text{ dBm}$$

Where:

- k = Boltzmann's constant = $1.38 \times 10^{-23} \text{ J/K}$
- T = Absolute temperature (Kelvin)
- B = Bandwidth (Hz)

Key Insights:

- Thermal noise power increases linearly with temperature and bandwidth.
- At room temperature ($T \approx 290\text{K}$) and 1 Hz bandwidth, noise power $\approx -174 \text{ dBm/Hz}$.
- It sets the fundamental lower bound on signal detection in receivers.



Watt $\rightarrow$ dB

$$P_{dB} = 10 \log_{10} \left(\frac{P_{\text{Watt}}}{1 \text{ W}} \right)$$

$$P_{\text{dBm}} = 10 \log_{10} \left(\frac{P_{\text{mW}}}{1 \text{ mW}} \right)$$

$$1 \text{ watt} \rightarrow 10 \log_{10} \left(\frac{1}{1} \right) = 0 \text{ dB}$$

$$1 \text{ watt} = 1000 \text{ mW} \rightarrow 10 \log_{10} \left(\frac{1000 \text{ mW}}{1 \text{ mW}} \right) = 30 \text{ dBm}$$

Noise for Real and Complex Modulation

Complex Up Conversion

$$S(t) = I + iQ$$

$$I \cos 2\pi f_c t$$

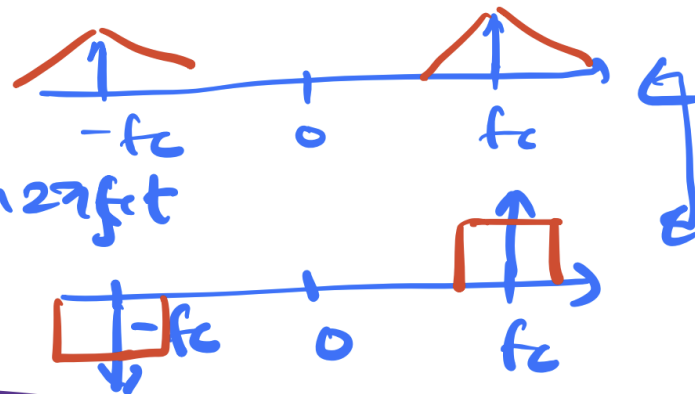
I(t)



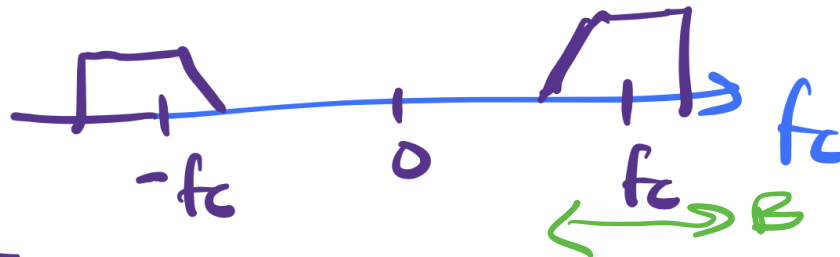
Q(t)



$$Q \sin 2\pi f_c t$$



$$I(t) \cos 2\pi f_c t - Q(t) \sin 2\pi f_c t$$

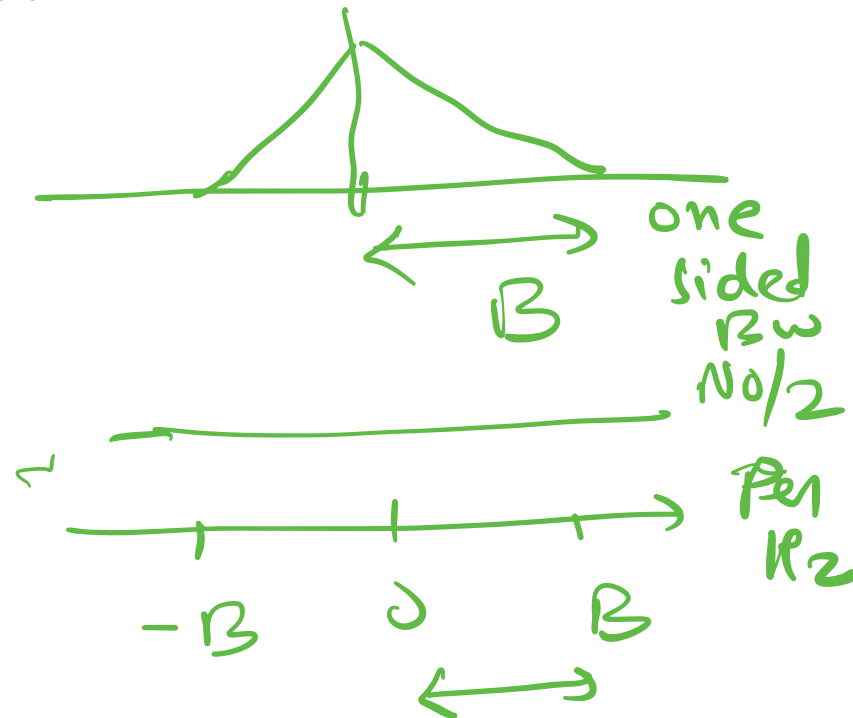


real signal
produce symmetric
magnitude freq response (ideally conj-symm.)

Noise for Real and Complex Modulation

$x(t)$ real $\rightarrow x(f)$

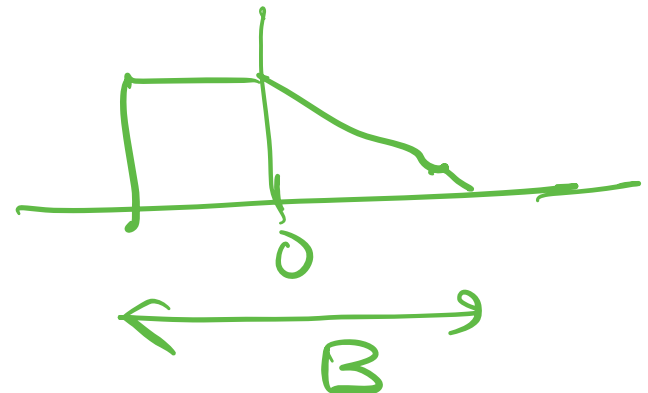
Noise $\rightarrow \left(\frac{N_0}{2} B\right)$ watt



Complex

Noise $\rightarrow \left(\frac{N_0}{2} B\right)$ watt

two sided BW



Decoding under noise

Assume Data: $y_i \sim \mathcal{N}(\theta, \sigma^2)$

$$= \theta + n \quad \begin{array}{l} \text{signal} \\ n \sim \mathcal{N}(0, \sigma^2) \end{array}$$

MLE (maximum likelihood):

$$\hat{\theta}_{MLE} = \arg \max_{\theta} p(\bar{\mathbf{y}} | \theta)$$

Chooses the parameter that makes the observed data most likely.

MAP (maximum a posteriori):

$$\hat{\theta}_{MAP} = \arg \max_{\theta} \underbrace{p(\theta | \mathbf{y})}_{\text{likelihood}} = \arg \max_{\theta} \underbrace{\log p(\mathbf{y} | \theta)}_{\text{likelihood}} + \log p(\theta)$$

Bayes rule

likelihood

prior

Maximum Likelihood Estimator

$$y_i \sim N(0, \sigma^2) \text{ iid}$$

$$\hat{\theta}_{MLE} = \max_{\theta} \log P(y_i | \theta)$$

↑ joint dist for all y_i

$$\max_{\theta} \log P(y_1 | \theta) \cdot P(y_2 | \theta) \cdots P(y_n | \theta)$$

$$\max_{\theta} \sum_i \log P(y_i | \theta)$$

$$\max_{\theta} \sum_i \log \left(\frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2} \frac{(y_i - \theta)^2}{\sigma^2}} \right)$$

$L(\theta)$

$$\underline{L(\theta)} = \sum_i \log\left(\frac{1}{\sqrt{2\pi}\sigma^2}\right)$$

$$\sum_i \left(-\frac{1}{2} \left(\frac{y_i - \theta}{\sigma^2} \right)^2 \right)$$

$$\frac{\partial}{\partial \theta} L(\theta) = \sum_i \frac{1}{\sqrt{2\pi}\sigma^2} 2(y_i - \theta) = 0$$

$$= \sum_i y_i = \sum \theta$$

$$\overset{n}{\theta_{MLE}} = \frac{1}{n} \sum_i y_i = n\theta$$

Properties of MLE estimator

(1) unbiased

$$E(\hat{\theta}_{MLE}) = \theta$$

$$E\left(\frac{1}{N} \sum y_i\right)$$

$$= \frac{1}{N} \sum E y_i = \frac{1}{N} \sum \theta$$

$$= \theta$$

(2) minimize variance
(asymptotically) large N

$$\text{Var}(\hat{\theta}_{MLE}) \rightarrow 0 \quad \text{for } N \rightarrow \infty$$

$$E\left((\hat{\theta}_{MLE} - \theta)^2\right)$$

$$= E\left(\left(\frac{\sum y_i}{N} - \theta\right)^2\right)$$

$$= \frac{1}{N^2} E\left[\left(\sum_i (y_i - \theta)\right)^2\right]$$

$$= \frac{1}{N^2} \sum_i E\left[\underbrace{(y_i - \theta)^2}_{\text{var } y_i = \sigma^2}\right]$$

$$= \frac{1}{N^2} \sum \sigma^2$$

$$= \frac{1}{N^2} N\sigma^2 = \frac{\sigma^2}{N}$$

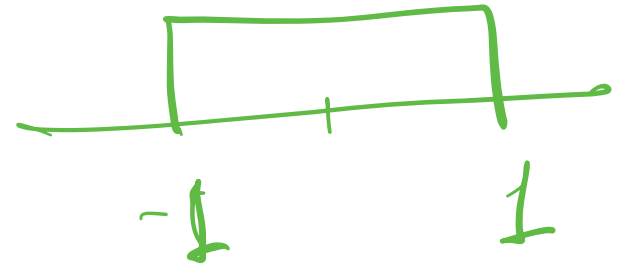
$$\rightarrow 0$$

$$\text{as } N \rightarrow \infty$$

Maximum A Posteriori Estimator

$$\max_{\theta} P(\theta | y)$$

$$\frac{\log P(y | \theta) \underbrace{P(\theta)}}{\downarrow}$$



Decision Boundary

$$y = s_m + \sqrt{}$$

- Decide between BPSK symbols $s \in \{\pm A\}$ given y .
- **ML detector**: $\arg \max_m p(y | s_m) \rightarrow$ nearest-neighbor rule.
- **MAP detector**: $\arg \max_m p(y | s_m) P(s_m) \rightarrow$ ML **plus** prior on symbol probabilities.
- Binary antipodal example $y = s + \mathcal{N}(0, \sigma^2)$, $s \in \{\pm A\}$:
Decide $+A$ if

$$\underbrace{\ln \frac{p(y | +A)}{p(y | -A)}}_{(2Ay)/\sigma^2} \geq \ln \frac{P(-A)}{P(+A)} \Rightarrow y \geq \frac{\sigma^2}{2A} \ln \frac{P(-A)}{P(+A)}.$$

$$(2Ay)/\sigma^2$$

priors.

- With equal priors the threshold is 0 $\rightarrow$ converges to ML.

Decision Boundary - MLE

$$y = \sum_{i=1}^n x_i + \nu \quad \nu \sim \mathcal{N}(0, \sigma^2)$$
$$x_i \in \{-A, A\}$$

$$P(y | -A) > P(y | A) \Rightarrow \text{decide } -A$$

$$\frac{P(y | -A)}{P(y | A)} > 1$$
$$= \frac{\frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(y - (-A))^2}{\sigma^2}}}{\frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(y - A)^2}{\sigma^2}}}$$

$$e^{-\left(\frac{y+A}{\sigma^2}\right)^2 + \left(\frac{y-A}{\sigma^2}\right)^2} > 1$$

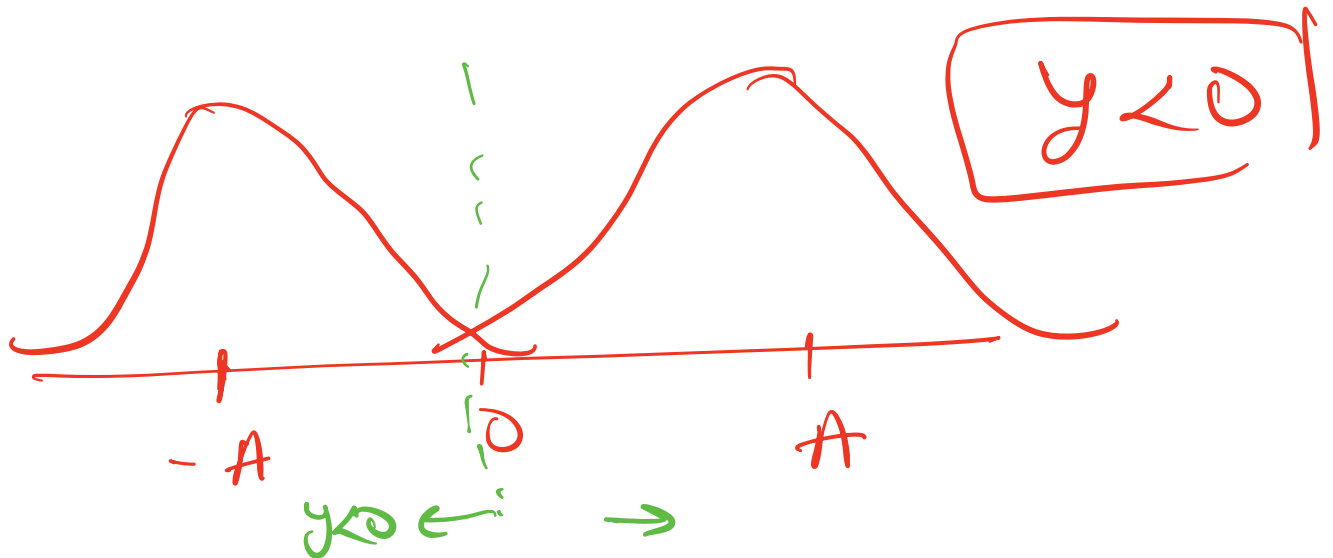
$$\frac{(y-A)^2 - (y+A)^2}{\sigma^2} > 0$$

$$\frac{a^2 - b^2}{\sigma^2}$$

$$\frac{(2y)(-2A)}{\sigma^2} > 0$$

$$\cancel{y} < \cancel{\frac{\sigma^2}{4A}}$$

$$-\frac{4Ay}{\sigma^2} > 0$$



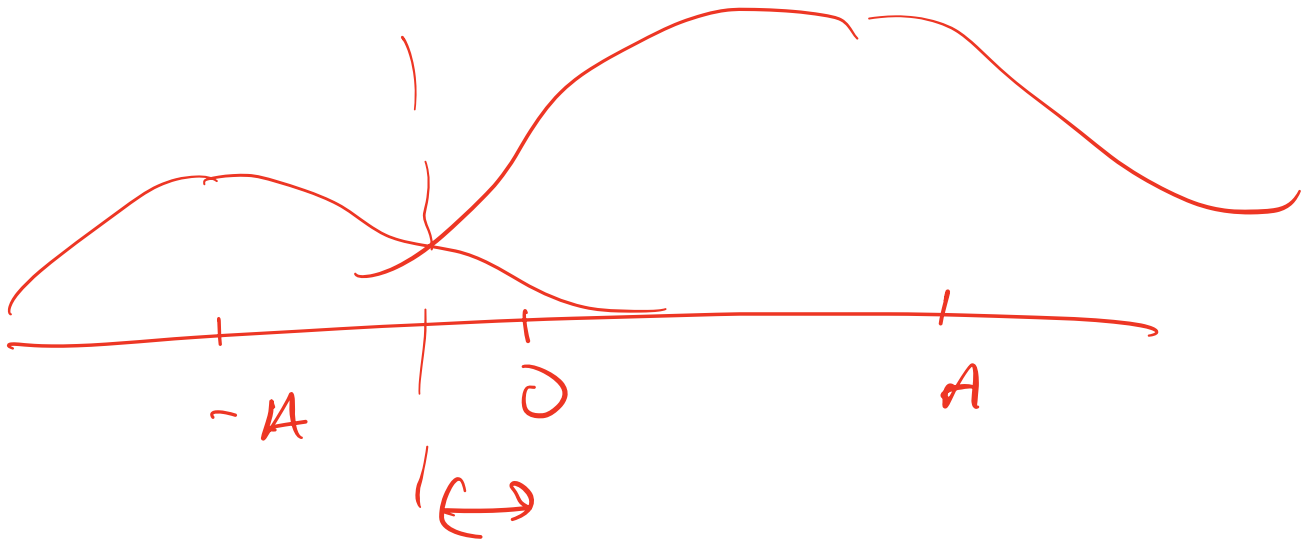
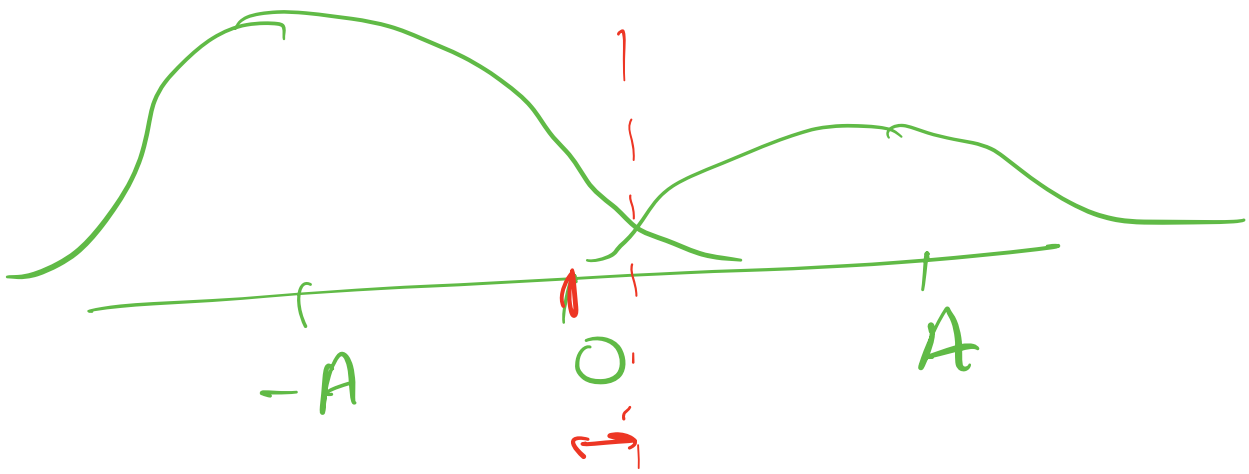
Decision Boundary - MAP

$$\frac{P(y|-A) P(-A)}{P(y|A) P(A)} > \overline{1}$$

$$\frac{P(y|-A)}{P(y|A)} > \frac{P(A)}{P(-A)}$$

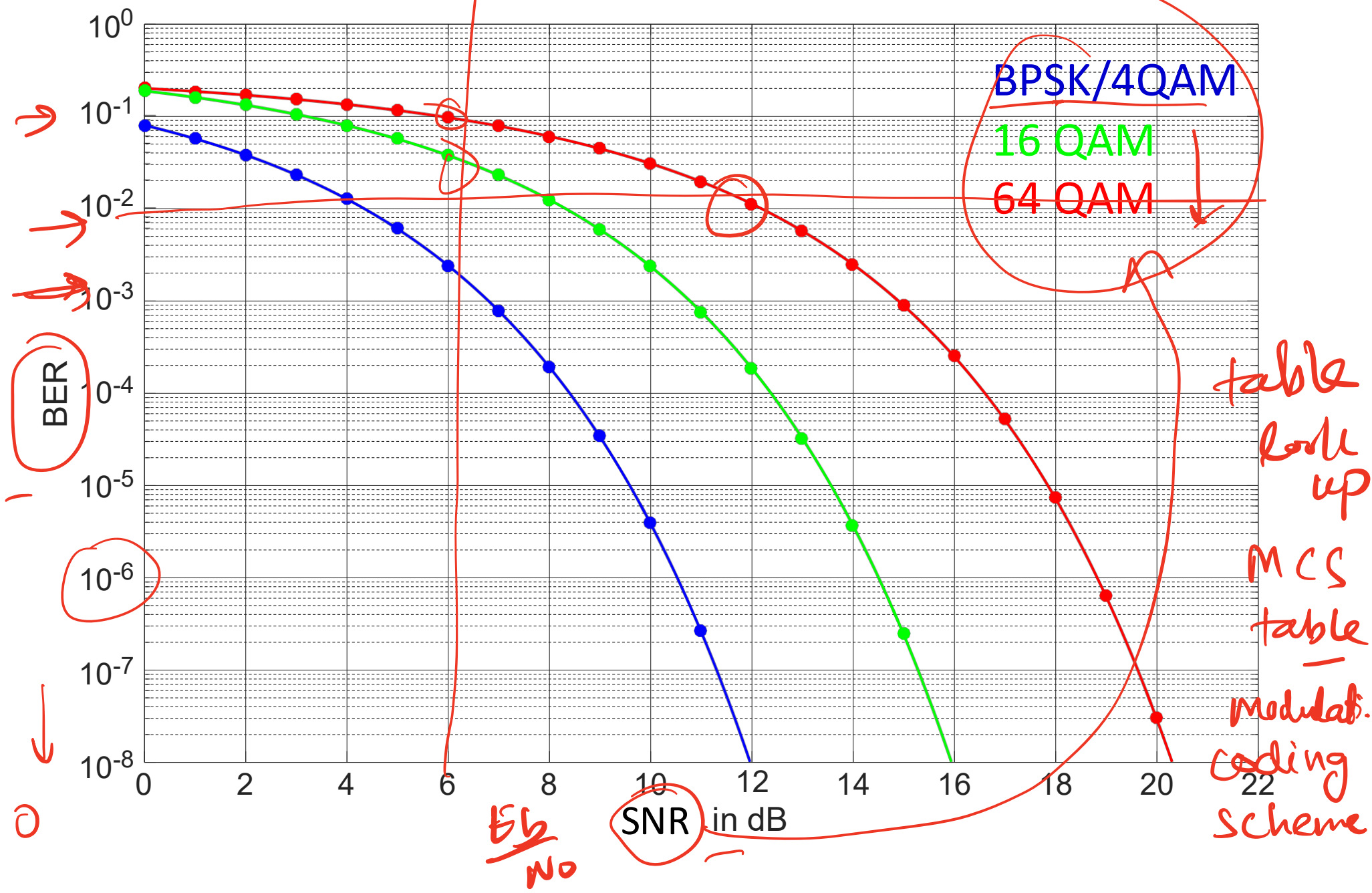
$$-\frac{4Ay}{\sigma^2} > \log \left(\frac{P(A)}{P(-A)} \right)$$

$$y < \frac{\sigma^2}{4A} \left[\frac{\log P(A) - \log P(-A)}{\log} \right]$$



Monte Carlo
Simulation.

Bit Error Rate (BER)



Bit Error Rate (BER)

$$y = s + v \in \mathcal{N}(0, \sigma^2)$$

Hypothesis $\in \{-A, A\}$ MLE

$$H_0 \quad -A \quad y < 0$$

$$H_1 \quad A \quad y > 0$$

bit error happen when

$$H_0 \quad y_i > 0$$

$$H_1 \quad y_i < 0$$

} error happen

$P(\text{bit error})$

$$y_i = -A + v$$

$$\begin{aligned} &= P(y_i > 0 | H_0) \text{ or } P(y_i < 0 | H_1) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(y_i - -A)^2}{\sigma^2}} = P(y_i | -A) \end{aligned}$$

$$P(-A + v > 0)$$

$$= P(v > A) = P\left(\frac{v}{\sqrt{\sigma^2}} > \frac{A}{\sqrt{\sigma^2}}\right)$$

$$\frac{v}{\sqrt{\sigma^2}} \sim N(0, 1)$$

Q function = $P(\text{st-normal dist} > a)$
Q(a)

$$P(v > A)$$

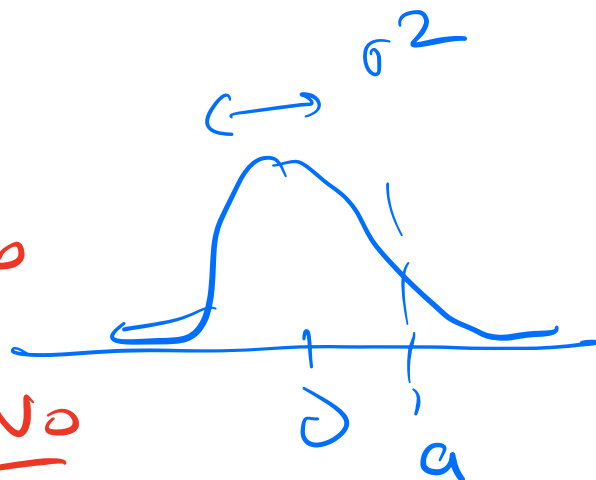
$$= Q\left(\frac{A}{\sqrt{\sigma^2}}\right)$$

$$= Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

$$A = \sqrt{E_b}$$

$$\sigma^2 = \frac{N_0}{2}$$

$$\text{where } \frac{E_b}{N_0} = \text{SNR}$$



Symbol Error Rate (SER)

16 QAM = $2^4 = 4$ ~~8~~ bits/symbol

bit error = P_b for each bit

what is symbol error rate?

SER

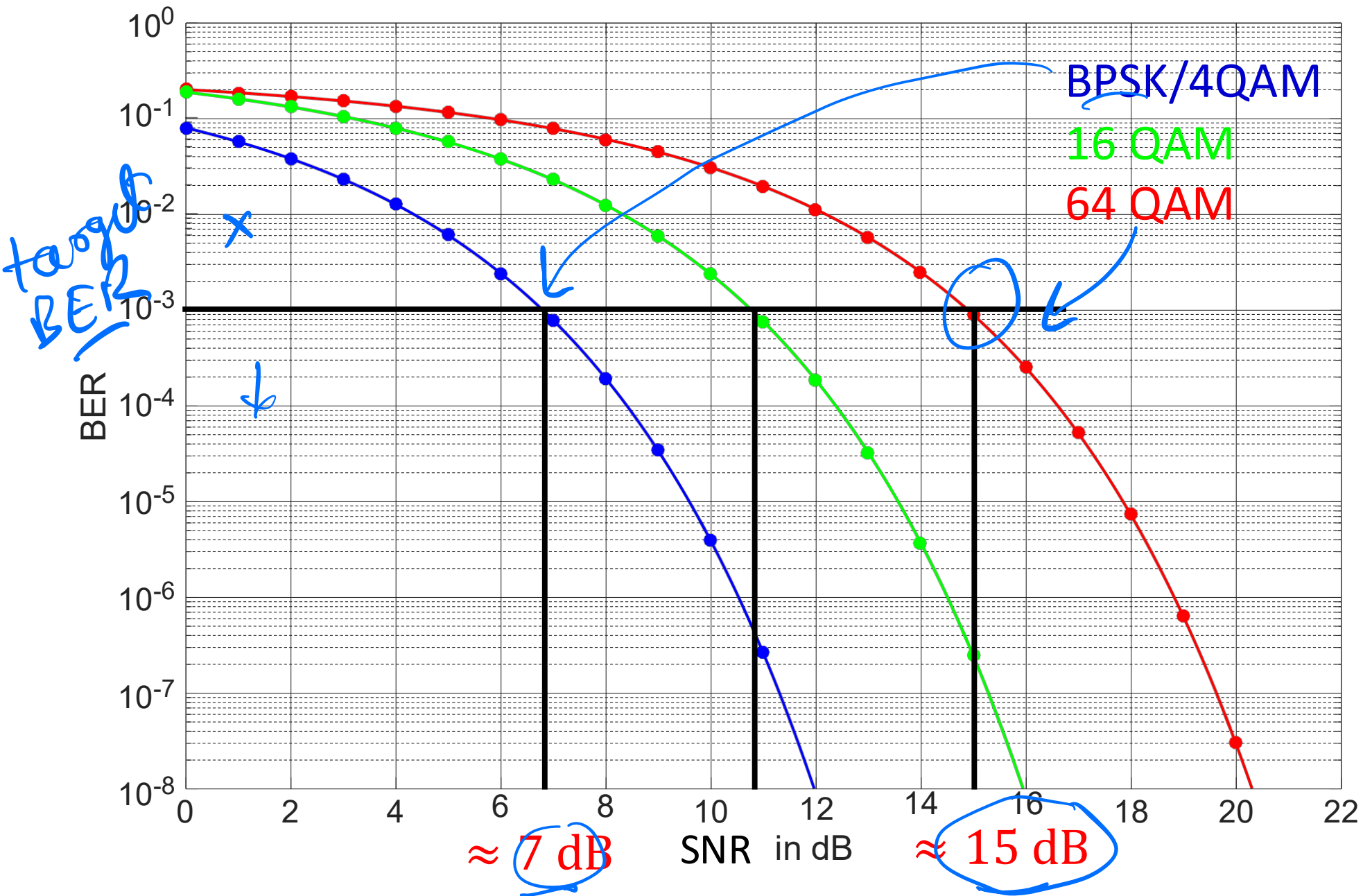
P_s = Prob that ~~any~~ one bit in error
at least

1 - no bits in error

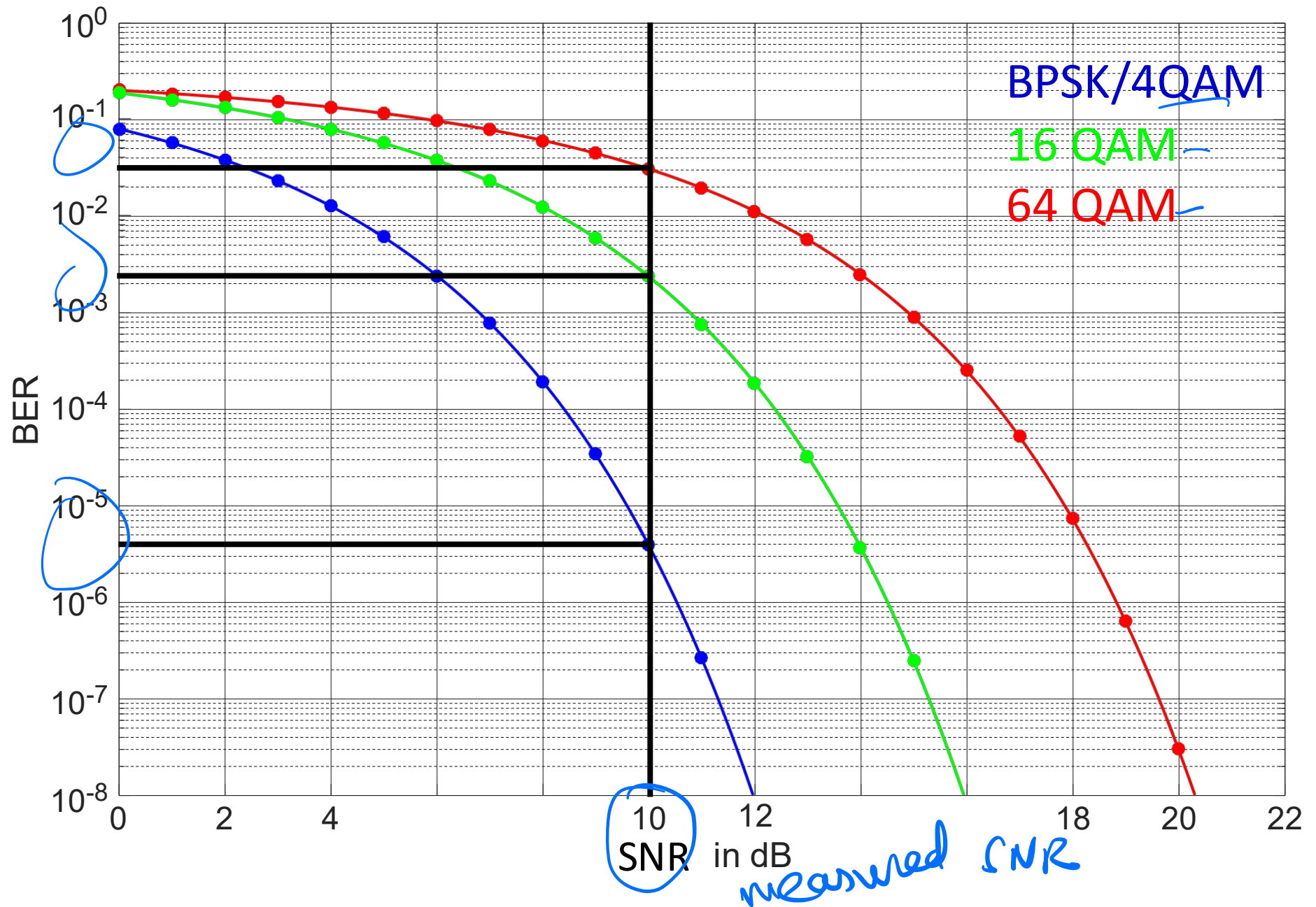
= 1 - bit 1 no error $(1 - P_b)$
x bit 2 no error $(1 - P_b)$

= $1 - (1 - P_b)^k$: k^{th} order mod.

Bit Error Rate (BER)



Bit Error Rate (BER)



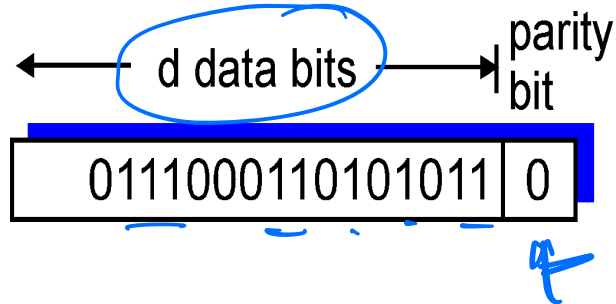
Error Detection

odd parity

- Add Redundant bits in order to detect errors at receiver

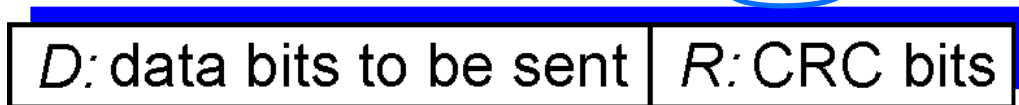
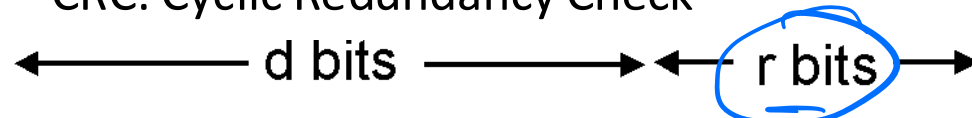
- Checksums → Detect Errors

- Parity Check



either 0 or 1 so that the whole $d+r$ block is odd (#1)

- CRC: Cyclic Redundancy Check



bit
pattern

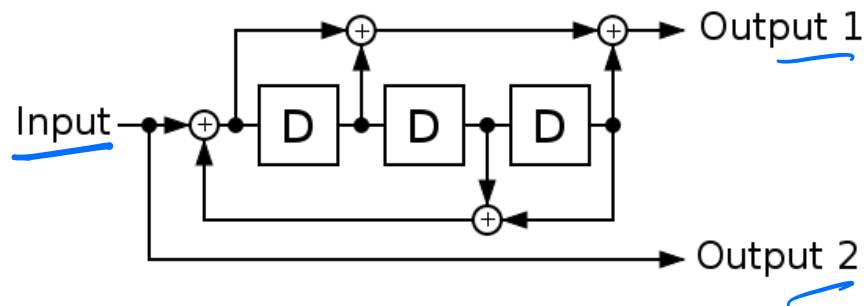
$d+r$
bits

- If packet (frame, sequence of bits) fails the checksum test
 - Packet is dropped
 - Packet loss.

Coding: Forward Error Correction (FEC)

- Add Redundant bits in order to correct errors at receiver
- FEC: Forward Error Correction

- Repetition Code
- Convolutional codes
- Reed Solomon codes
- Turbo codes
- Polar codes
- LDPC codes



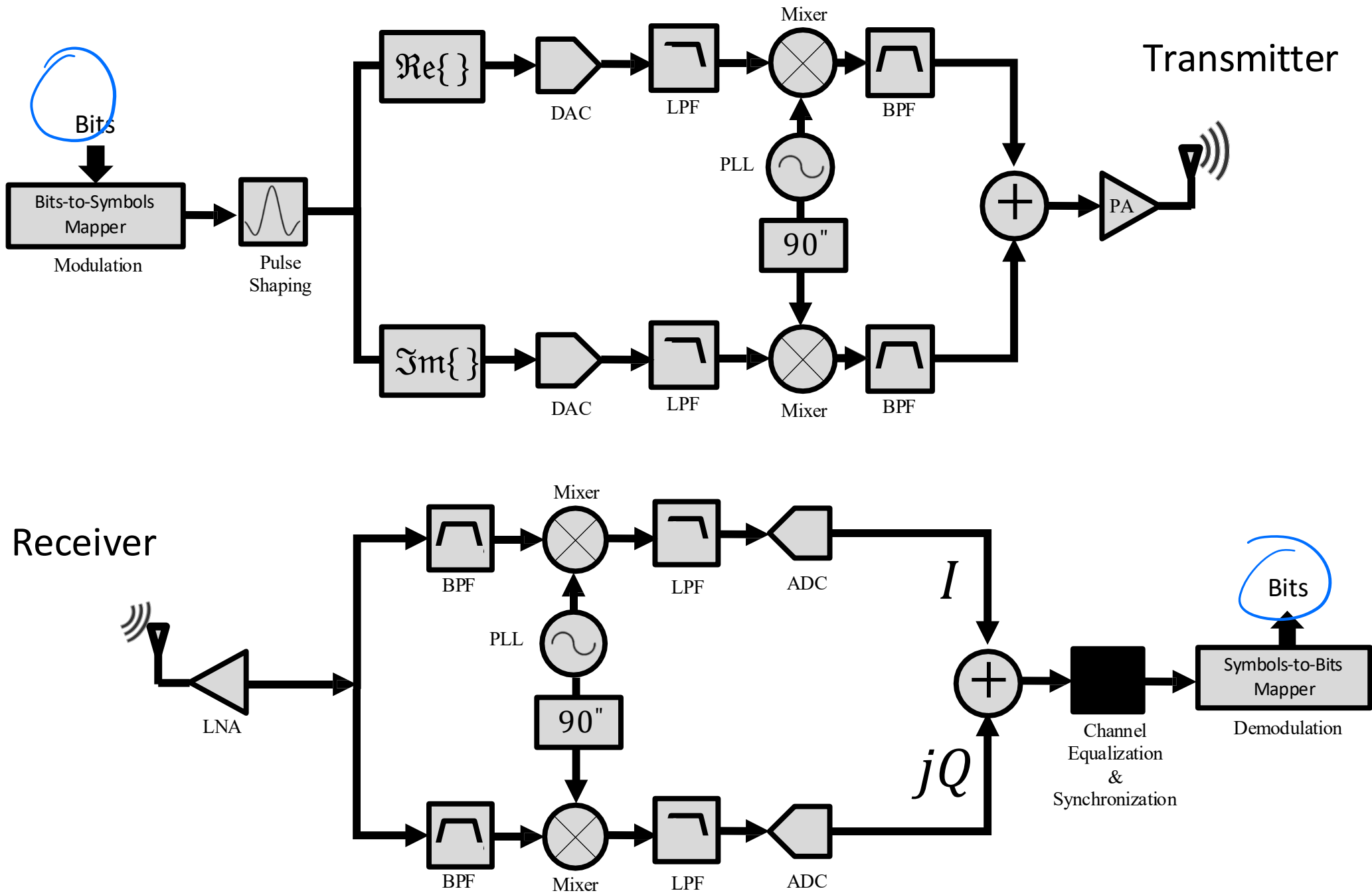
- Code Rate =
$$\frac{\# \text{ of data bits (uncoded)}}{\# \text{ of transmitted bits (coded)}}$$

- E.g. coding rate $\frac{2}{3}$, for every 2 data bits, I transmit 3 bits.

$\frac{2}{3} < 1$

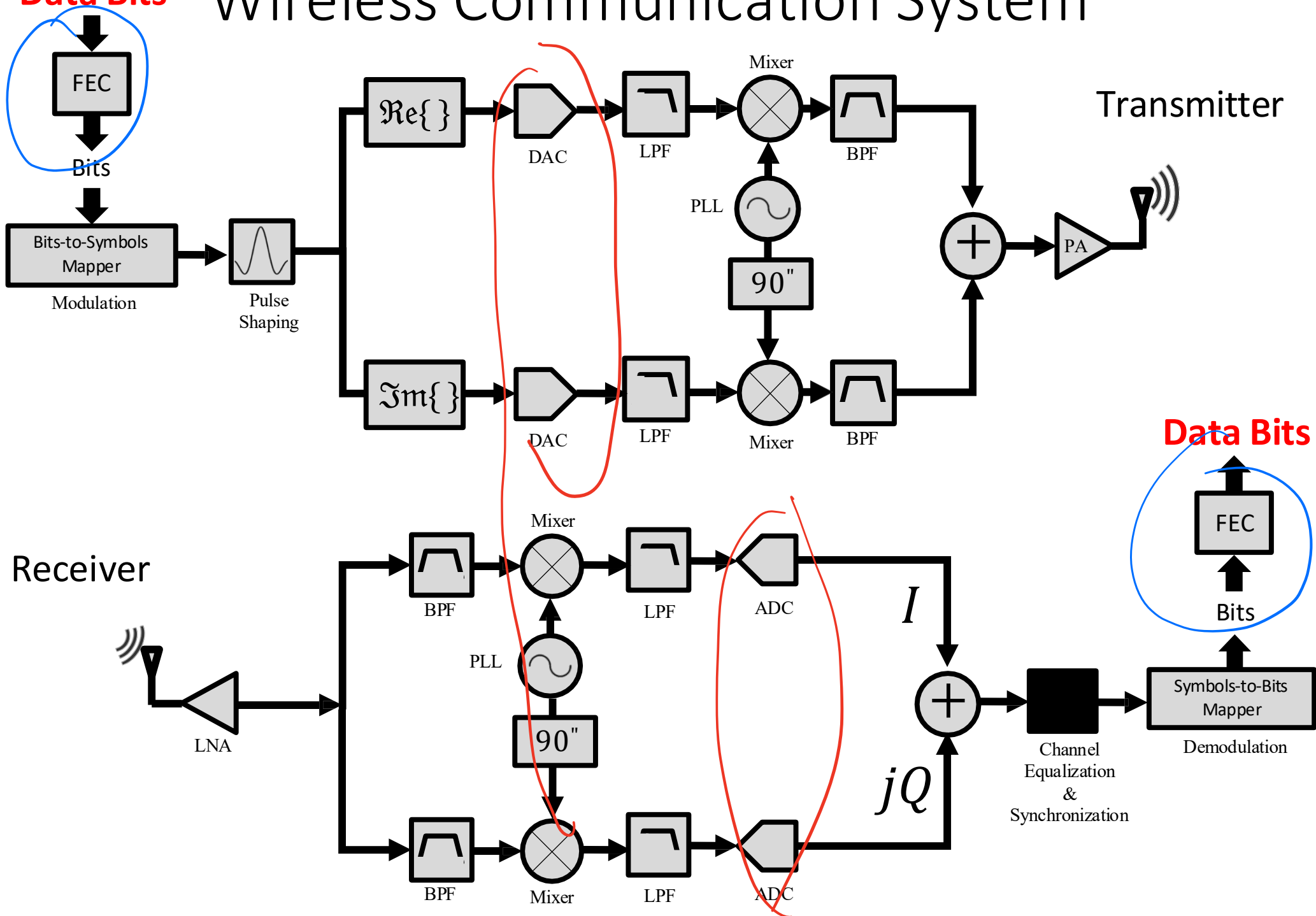
$2 + 1$

Wireless Communication System



Data Bits

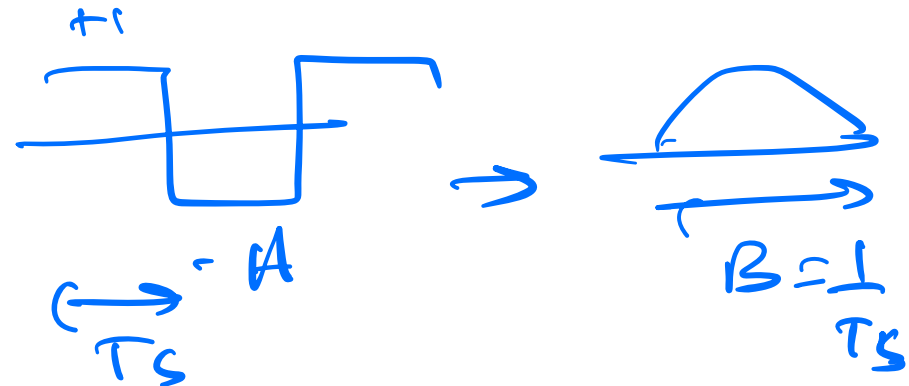
Wireless Communication System



Wireless Data Rates

- **Data Rate**

- Bandwidth: Samples/sec
- Modulation: Bits/sample
- Coding Rate: Data Bits/Coded Bits



$$\propto \underbrace{\text{Data Rate}}_{\text{bits/sec}} = \underbrace{\text{Bandwidth}}_{\text{bits/sec}} \times \underbrace{\text{Bits/sample}}_{\text{modul.}} \times \underbrace{\text{Code Rate}}_{\text{code}}$$

- **Capacity**

- Maximum Achievable Data Rate
- Shannon Capacity Theorem:

$$BER \approx 0$$

$$\underbrace{\text{Capacity}} = \underbrace{\text{Bandwidth}} \times \underbrace{\log_2(1 + \text{SNR})}$$